

Runtime adaptation as a programming pattern in service composition



Carlos G. Lopez Pombo@



Pablo Montepagano@



Emilio Tuosto@



Coordination
Urbino 10 May, 2026

Take-away message

Take-away message

Runtime architectural adaptation

Take-away message

Runtime architectural adaptation
= Monitoring

Take-away message

Runtime architectural adaptation

= Monitoring

+ Decision policy

Take-away message

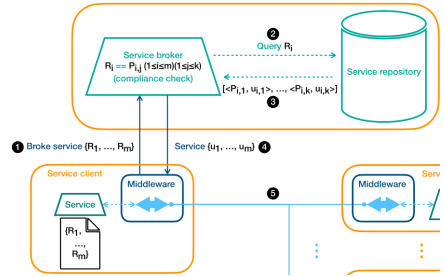
Runtime architectural adaptation

= Monitoring

+ Decision policy

+ Semantic-based service composition

from



Take-away message

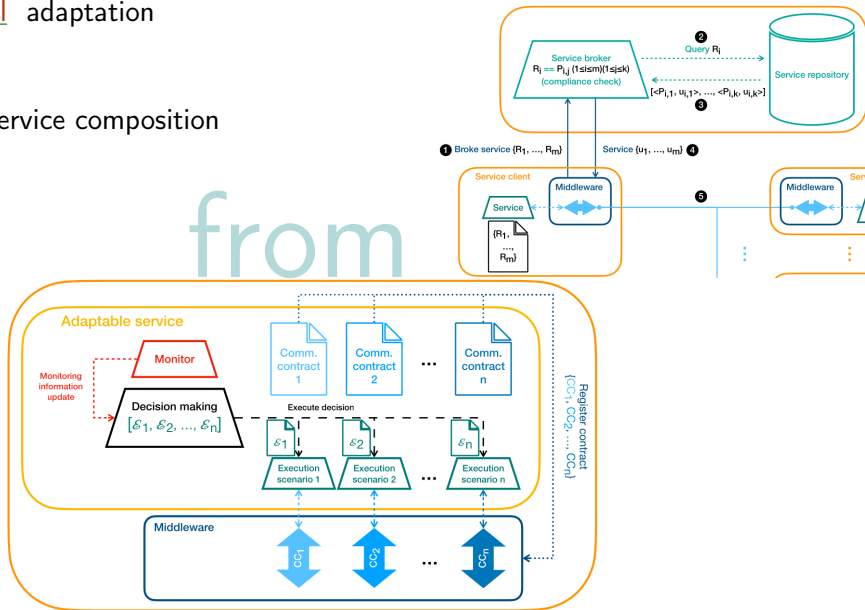
Runtime architectural adaptation

= Monitoring

+ Decision policy

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from
to



- ▶ A bird-eye watch of *SEArch*

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- ▶ Adaptation (as we see it)

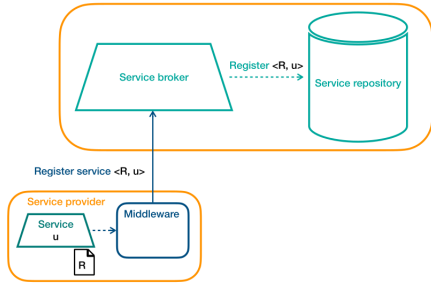
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- ▶ Our programming pattern

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- ▶ Some conclusions

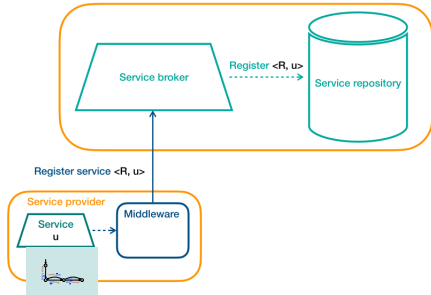
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– Prelude –

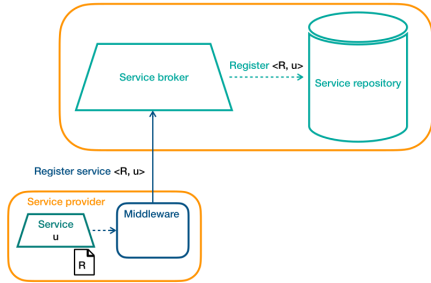
SEARCH basics



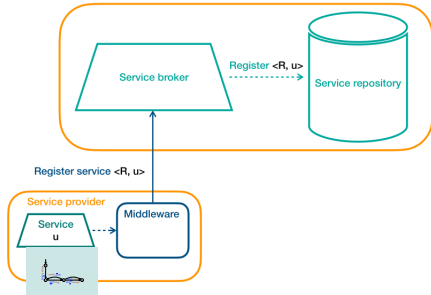
SEArch basics



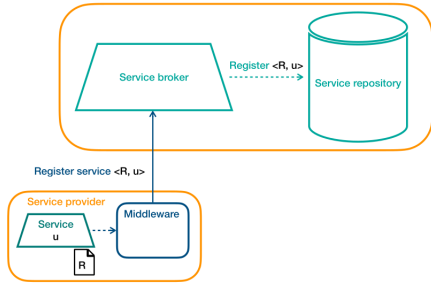
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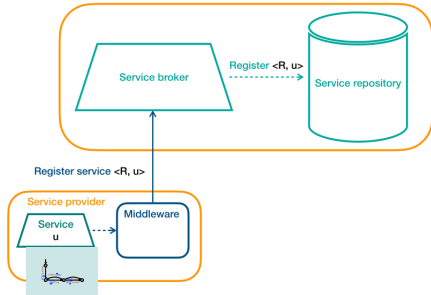
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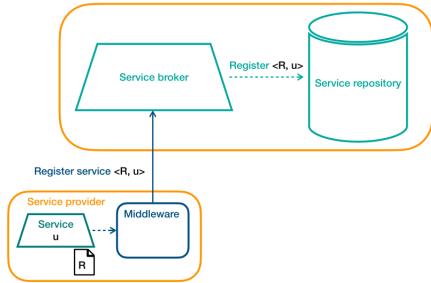
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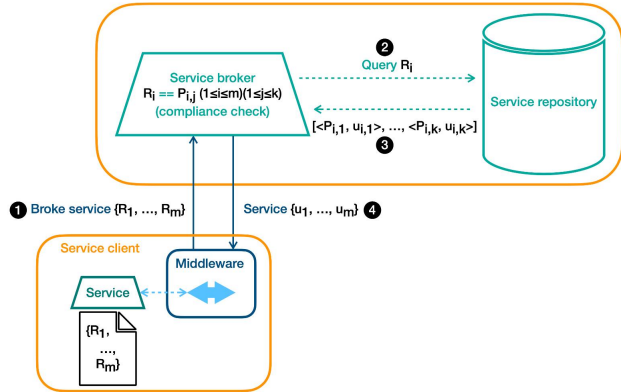
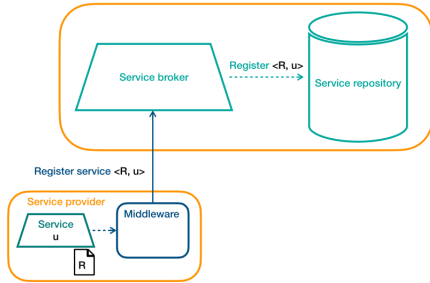
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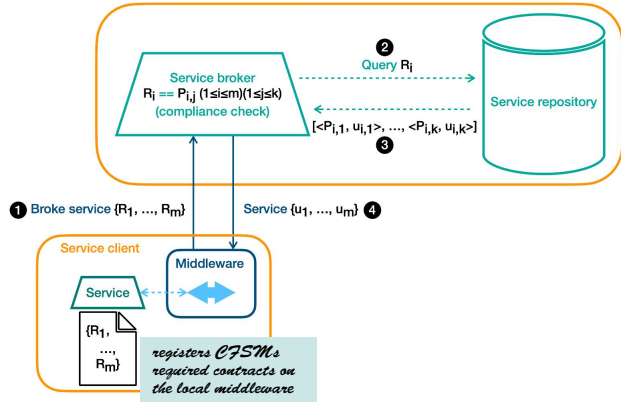
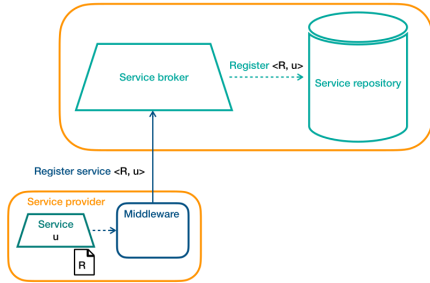
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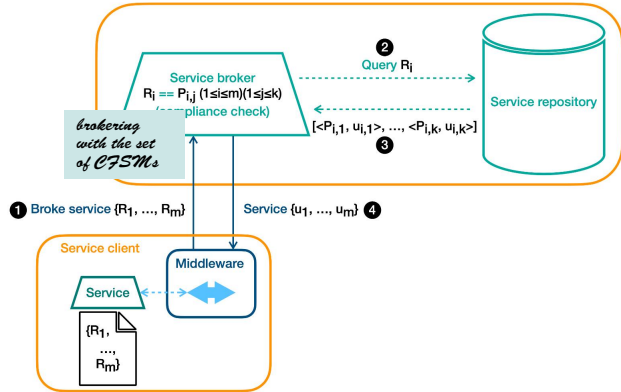
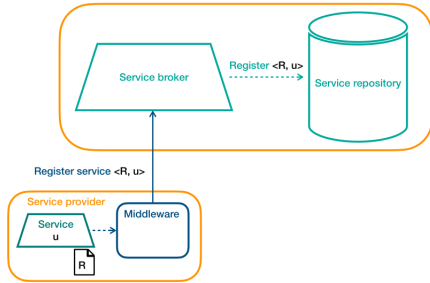
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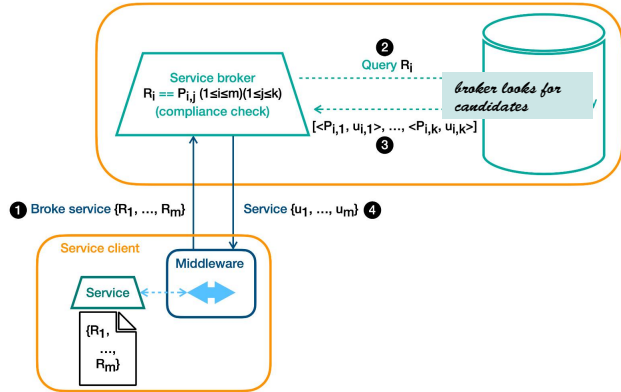
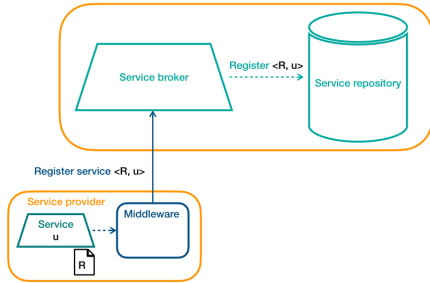
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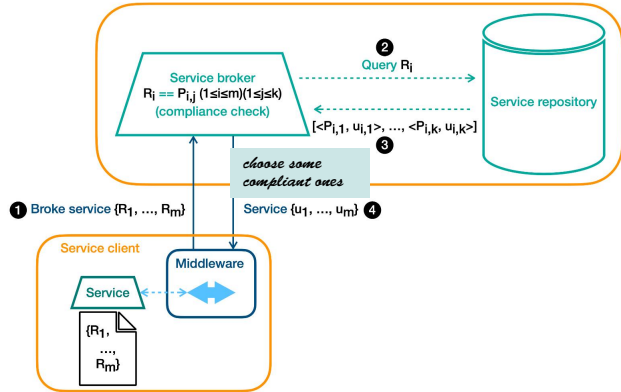
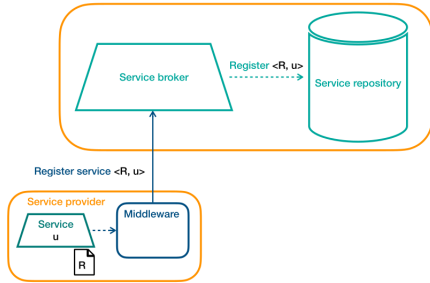
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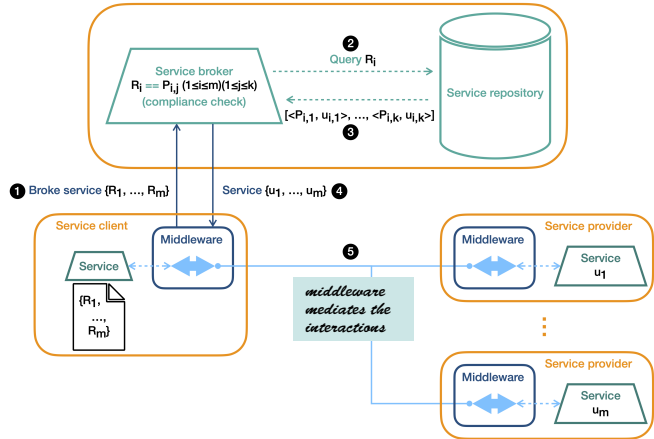
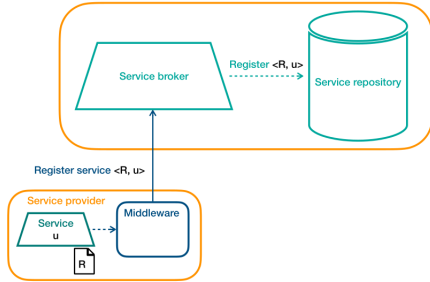
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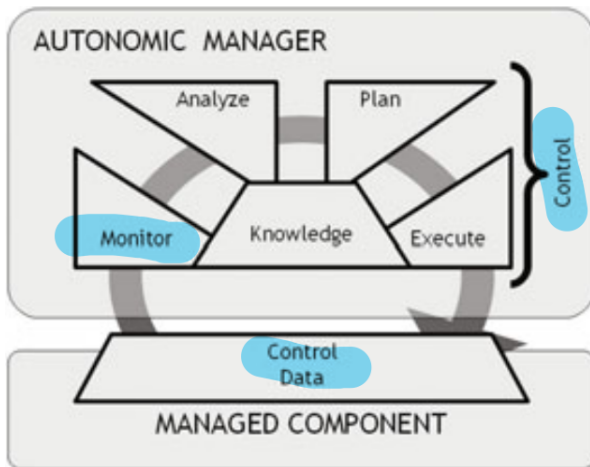


SEARCH basics



– On adaptation –

Adaptation basics



Components' behaviour = control + data

- ▶ *adaptation = process/modify specific computational/physical resources*
- ▶ *control data [?]^a*
 - ▶ *memory usage*
 - ▶ *network traffic*
 - ▶ *room temperature*
 - ▶ *...*

*Control data are an
"adaptation"-dependent design choice*

^aPicture borrowed from [?]

A benchmark for adaptation

Bliudze et al.

5

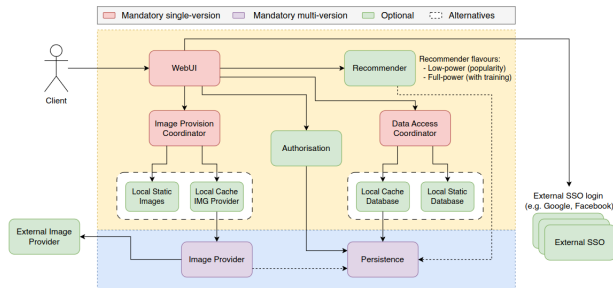


Figure 2: Adaptable TeaStore Architecture (functionalities shown within the coloured areas are internal, those in the blue area are outsourceable, in the yellow area—non-outsourceable (cf. Table 1))

“a broad catalogue of reference adaptation scenarios centred around Adaptable TeaStore, useful to evaluate the ability of a given adaptation technology to address conditions such as component unavailability, cyberattacks, provider outages, benign/malicious traffic increases, and user-triggered reconfigurations” [?]

A running example

¹Adapted from [?, ?]

A running example

The adaptable TeaStore benchmark¹

- ▶ a webUI (app users' entry point)
- ▶ a remote image provider
- ▶ a locally hosted DB of images

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A running example

The adaptable TeaStore benchmark¹

- ▶ a webUI (app users' entry point)
- ▶ a remote image provider
- ▶ a locally hosted DB of images

According to the specification

- ▶ providers process images (e.g., applying filters, resizing, etc.);
- ▶ webUI automatically generates webpages with images
 - ▶ from image providers
 - ▶ or from DB for runtime adaptation

¹Adapted from [?, ?]

Our adaptable TeaStore

Our adaptable TeaStore

Adaptation triggered by service degradation depending on

- ▶ network Latency
- ▶ Responsiveness of image provider
- ▶ CPU load of webUI

$$\mathcal{E}_1 = \forall x : L < x \implies R < x^2$$

$$\mathcal{E}_2 = \neg \mathcal{E}_1 \quad \wedge \quad 0 \leq C \leq .9$$

$$\mathcal{E}_3 = \neg \mathcal{E}_1 \quad \wedge \quad C > .9$$

Our adaptable TeaStore

Adaptation triggered by service degradation depending on

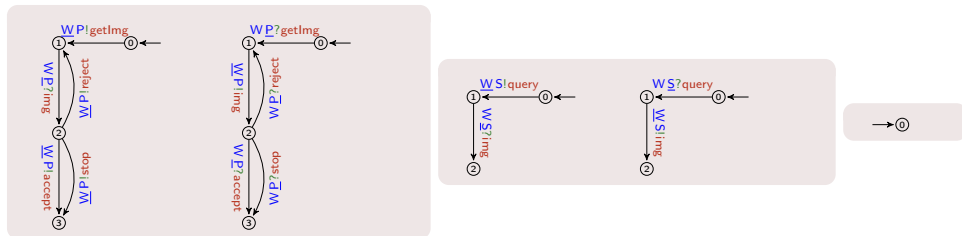
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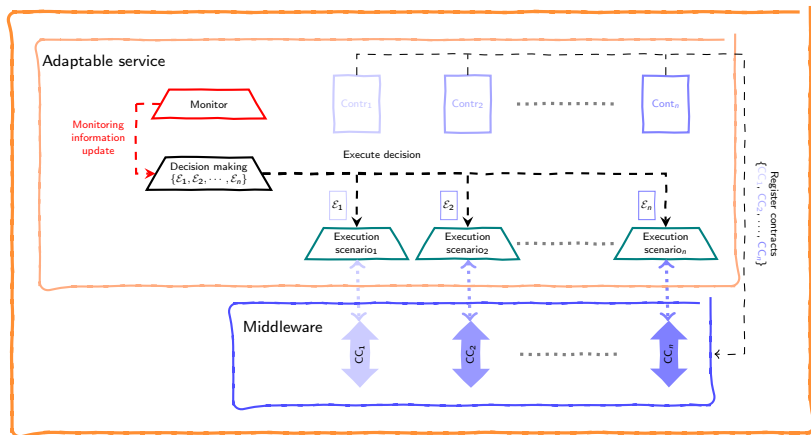
$$\mathcal{E}_3 = \neg \mathcal{E}_1 \quad \wedge \quad C > .9$$

Contracts for the 3 scenarios:

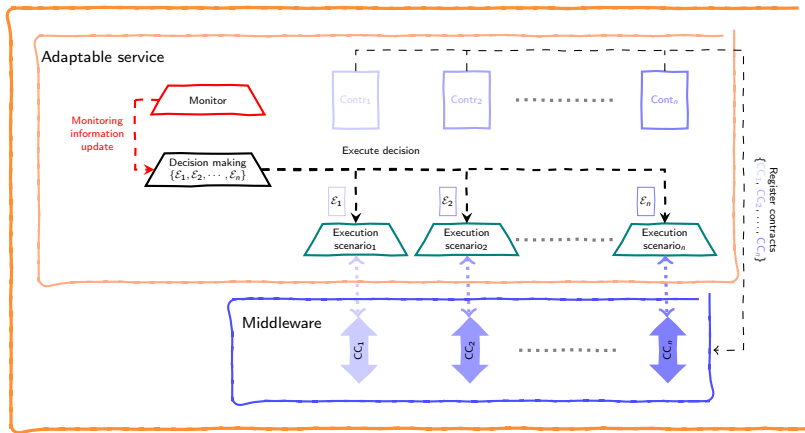


– Adaptation as service composition –

Architecture of our adaptation pattern



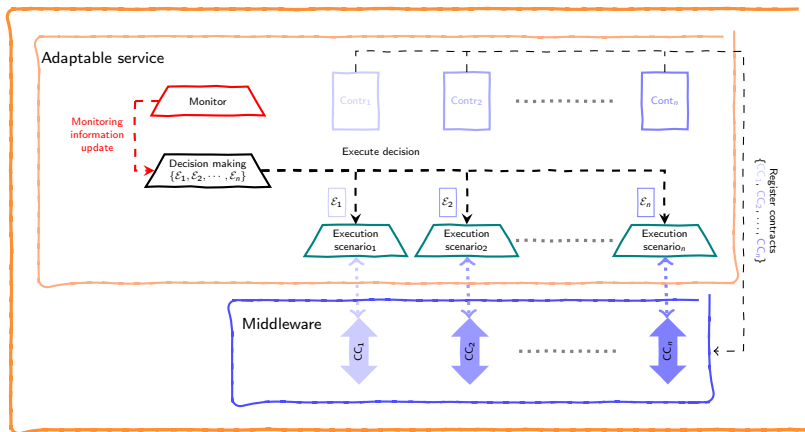
Architecture of our adaptation pattern



Middleware

- ▶ mediates interactions
- ▶ through ports' connections

Architecture of our adaptation pattern



Adaptable service

- ▶ a contract for each channel
- ▶ execution scenarios imp.
- ▶ adaptation policy

Middleware

- ▶ mediates interactions
- ▶ through ports' connections

Control loop

while ...:

m = monitor's last update of control data

next_scenario = choose(m)

if next_scenario $\neq$ running_scenario:

stop running_scenario

trigger next_scenario

$$m : \begin{cases} L \mapsto 2 \\ R \mapsto 3 \\ C \mapsto .5 \end{cases}$$

and e.g., rely on *RCF* [?, ?, ?]
which uses a *SM?* solver

A control loop as “service combinator”

```
1 from threading import Thread
2 from ExcScenarios import ES, done
3
4 done = False
5
6 while <not done>:
7     stop, scenario = False, None
8     i = 0
9     while scenario == None:
10         m = monitor_update()
11         if not SMT(not  $\mathcal{E}_i[a \mapsto m(a)]_{a \in \mathcal{A}}$ ) == SAT:
12             scenario = Thread(target = ES[i])
13             scenario.start()
14         else:
15             i = (i + 1) mod len(ES)
16
17 while not (done or stop):
18     old_m = m
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ES *list of functions each implementing
an execution scenario*

done *shared for coordinated termination*

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*spawn the chosen scenario ES[i],
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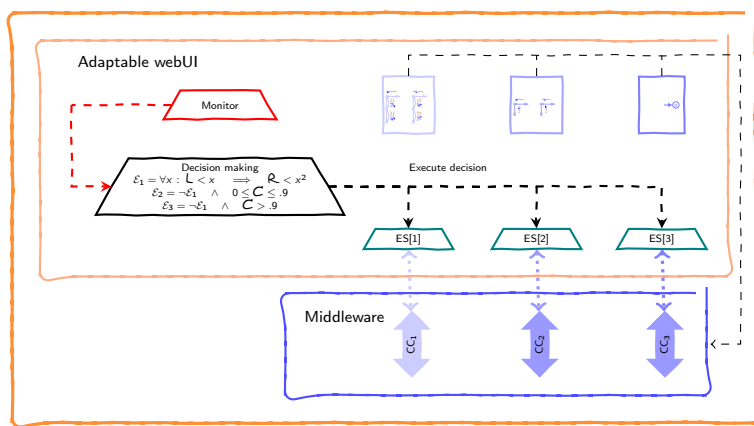
get control data

*spawn the chosen scenario $ES[i]$,
if enabled*

*keep going until the scenario is done or
adaptation is needed*

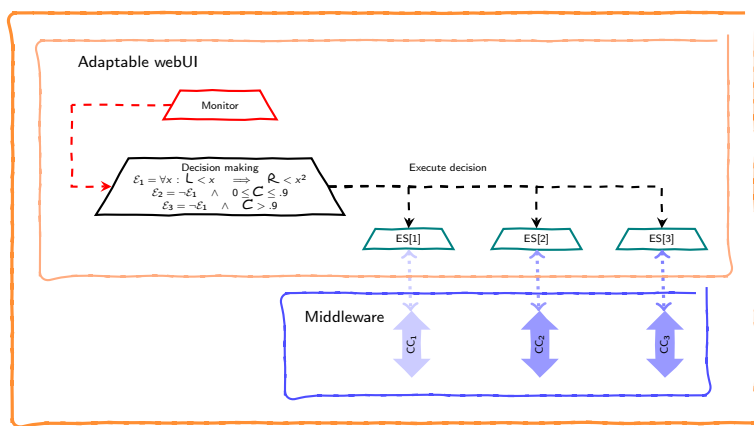
– Our pattern at work –

The pattern in action

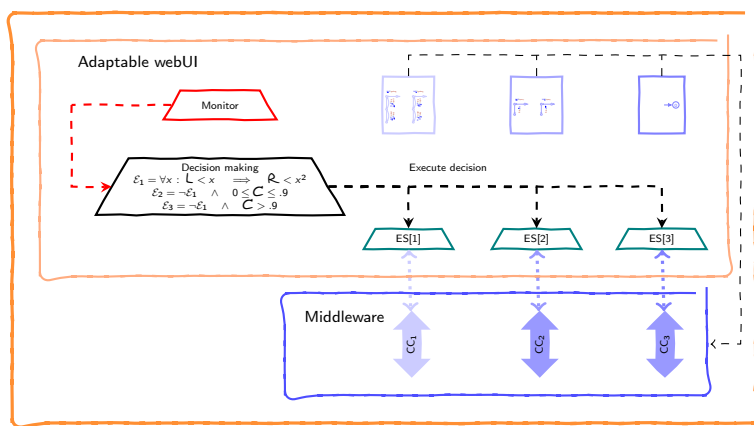


*Comm. contracts registered
on local middleware*

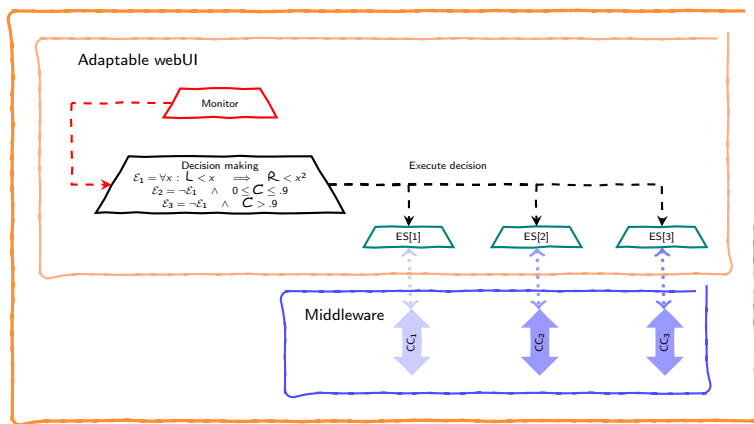
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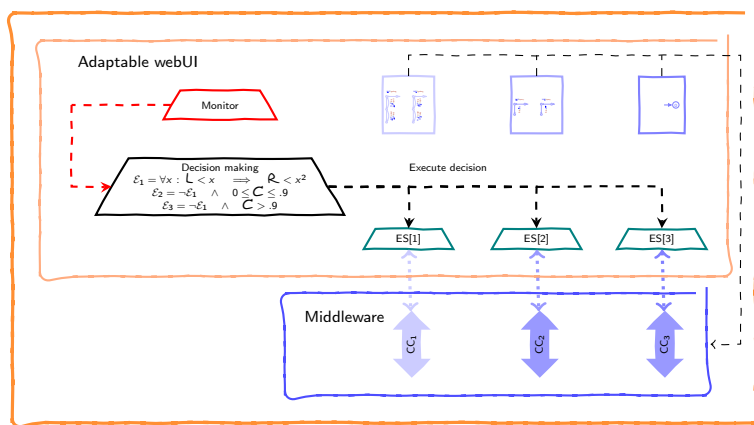
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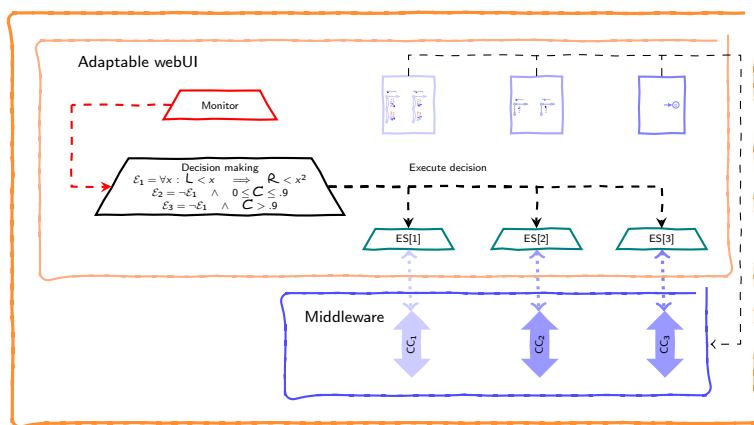


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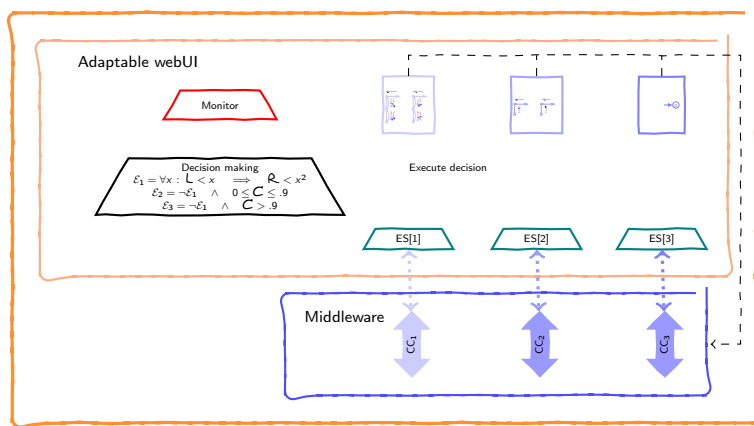
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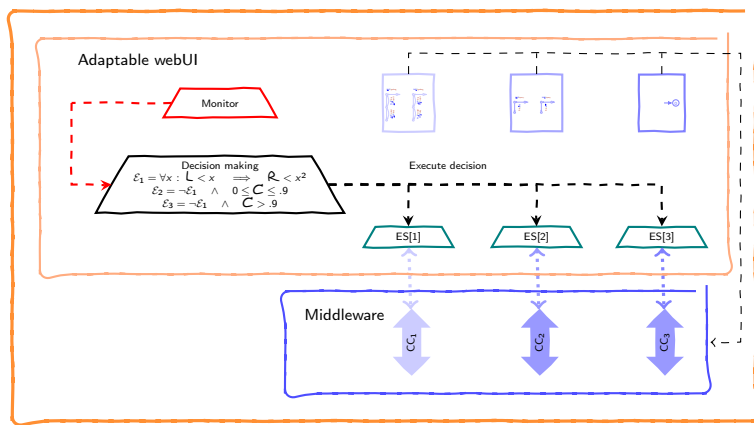
Monitor and decision procedure spawned

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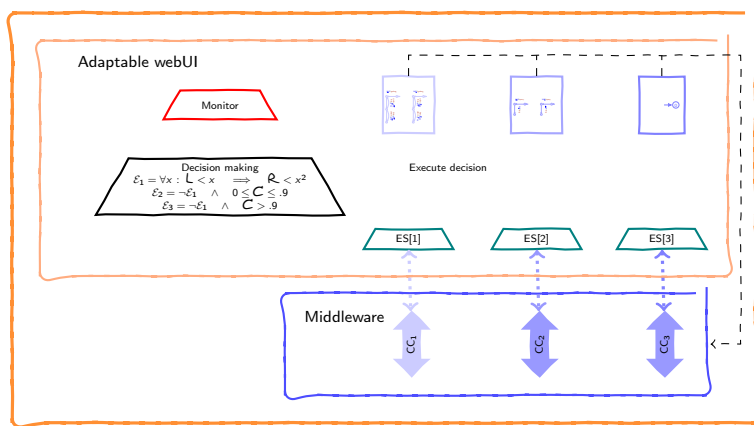
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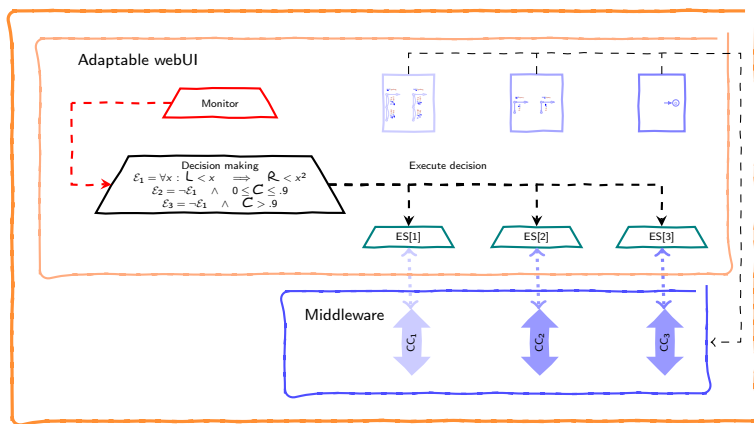
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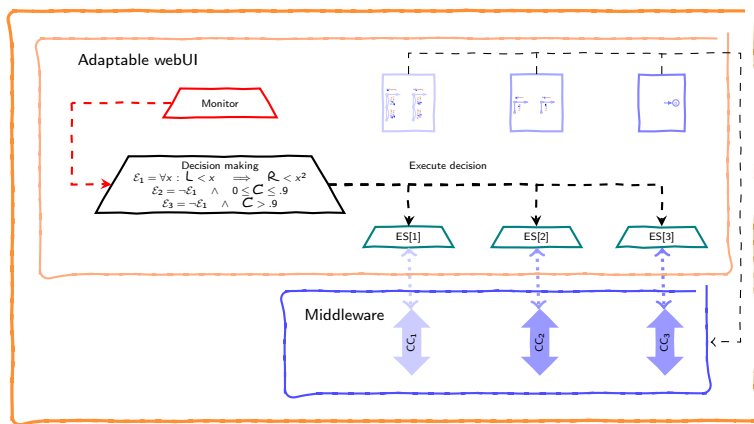
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Control data acquired:
 $L \mapsto 2, R \mapsto 3, C \mapsto 0.54$

The pattern in action



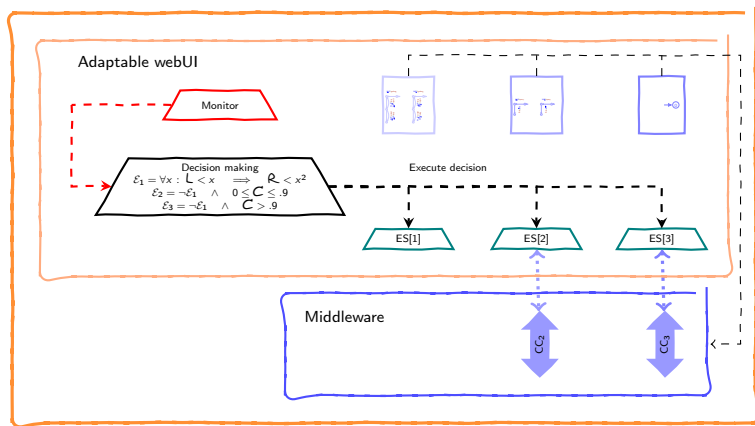
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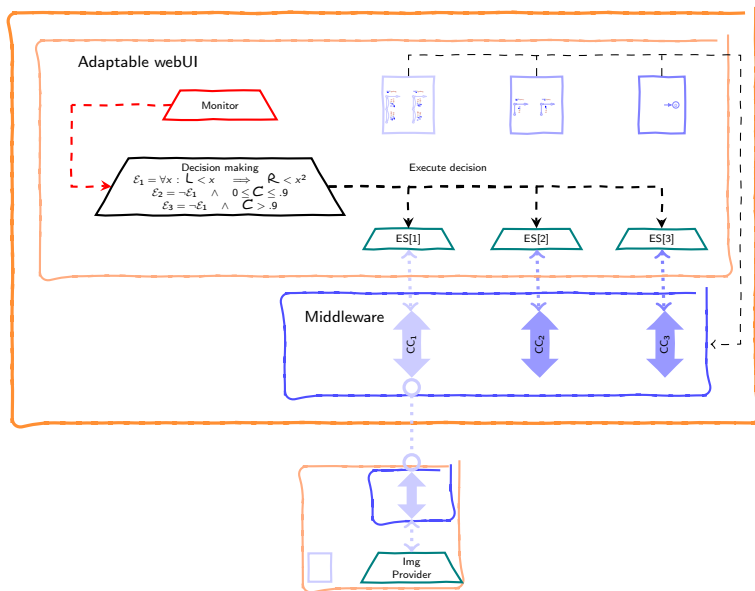


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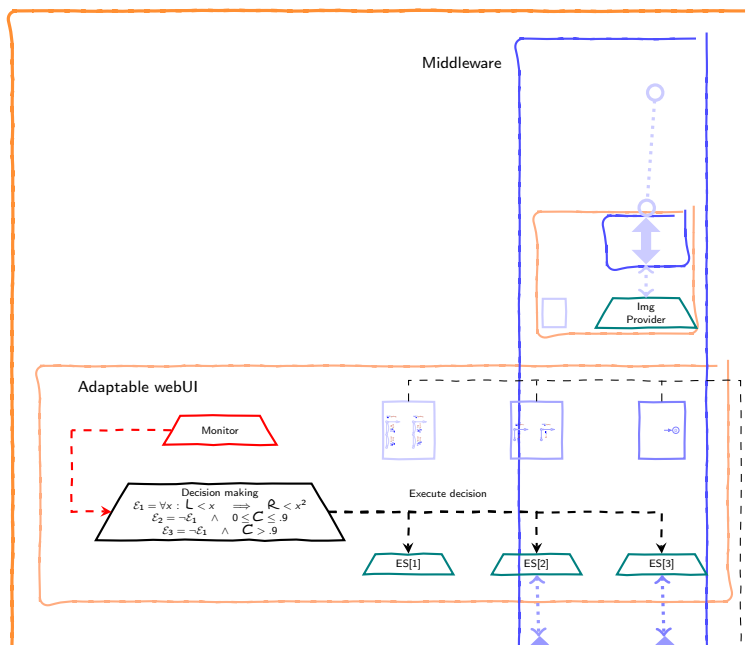
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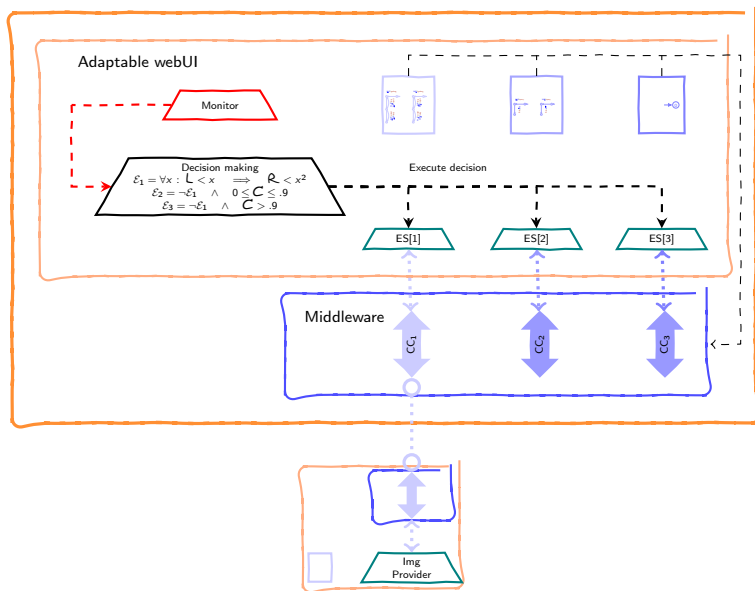
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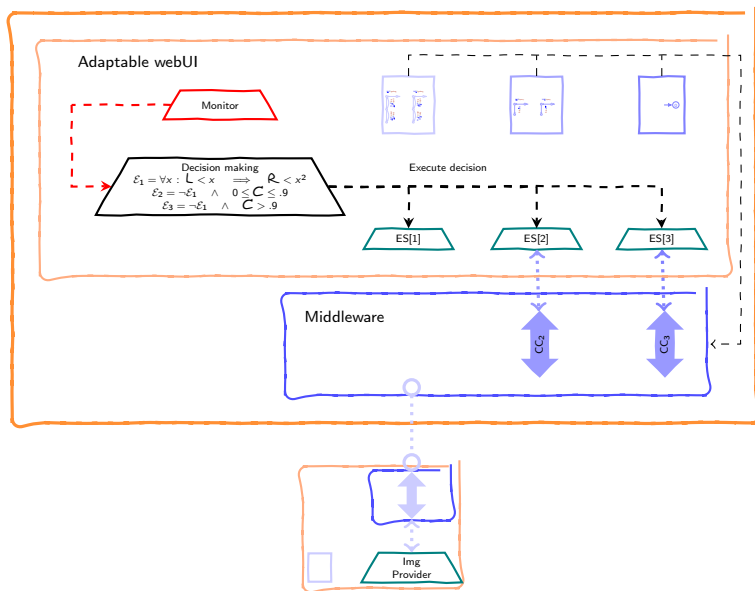
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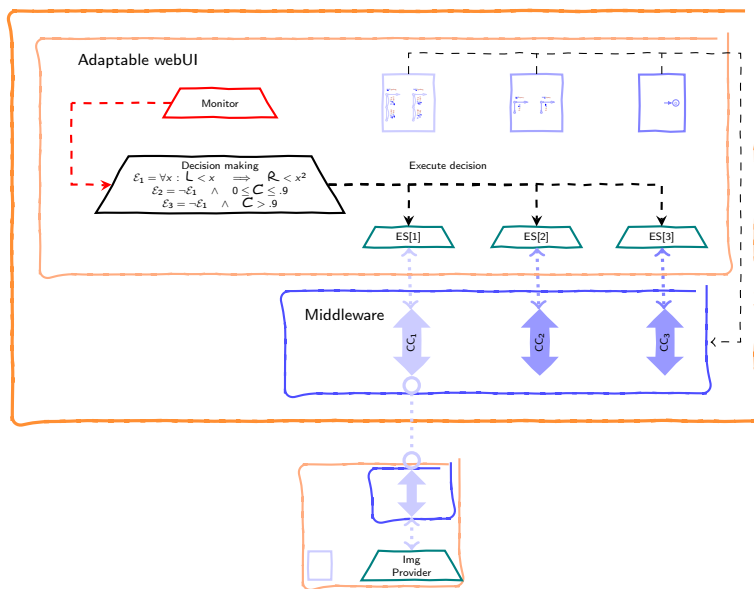
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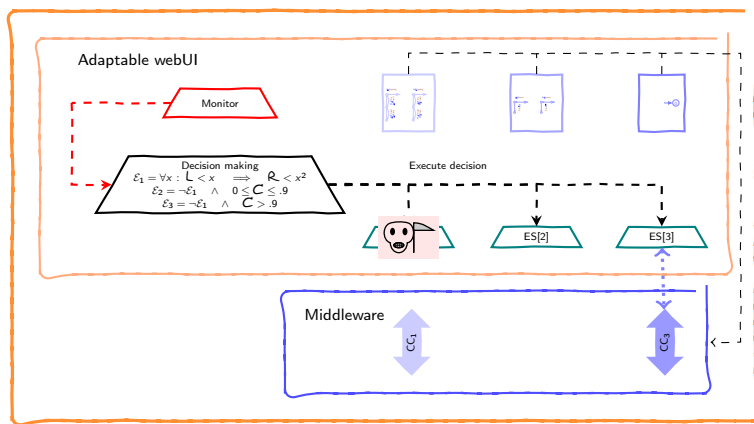


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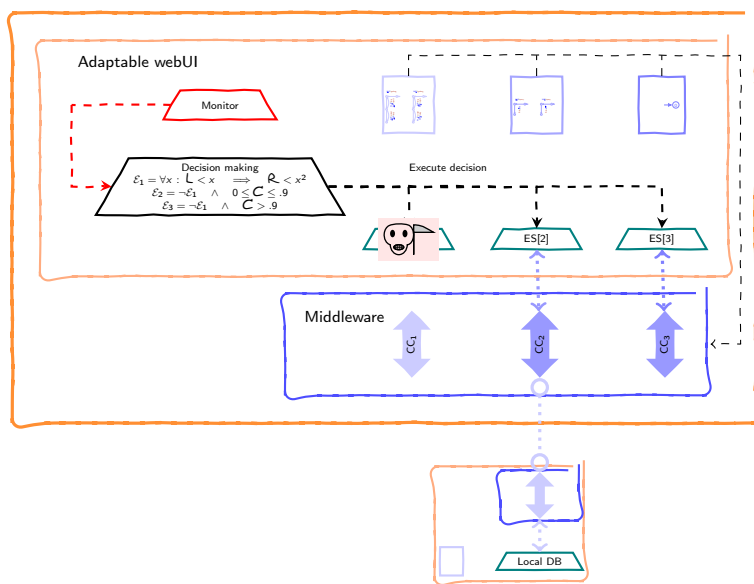
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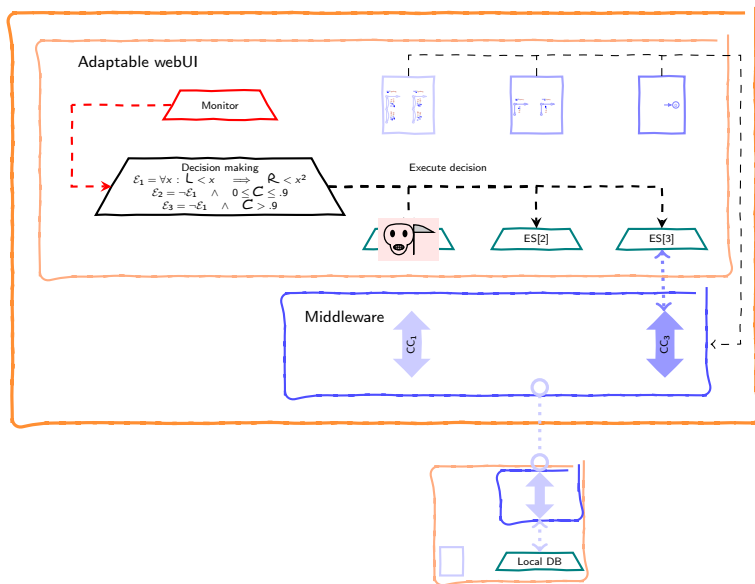
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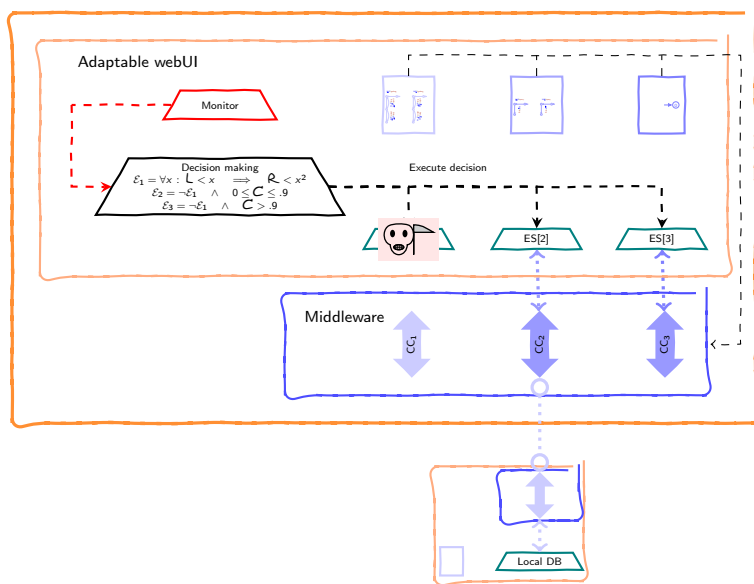
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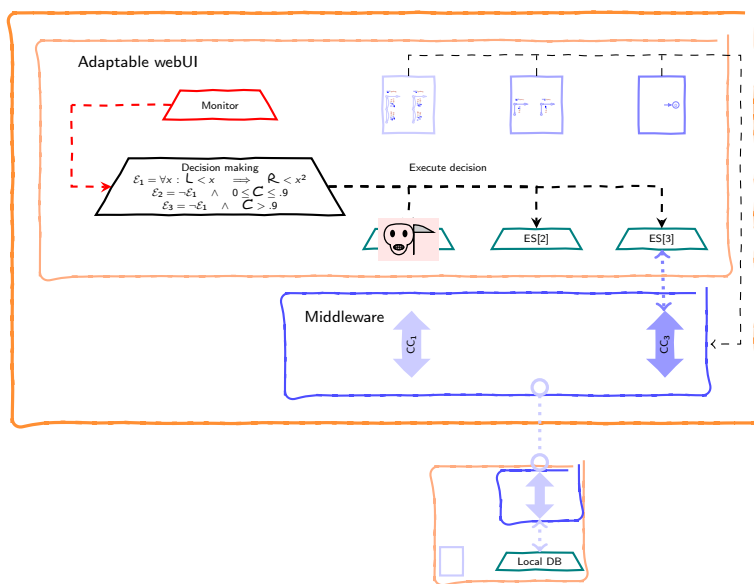
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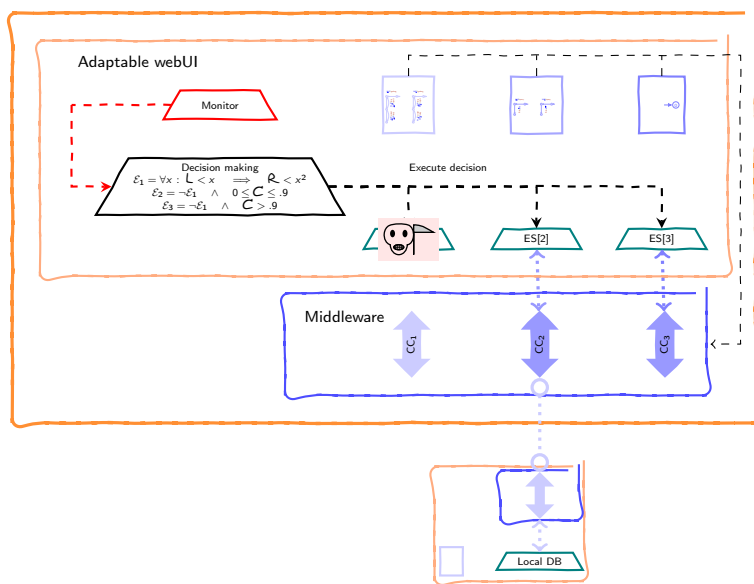


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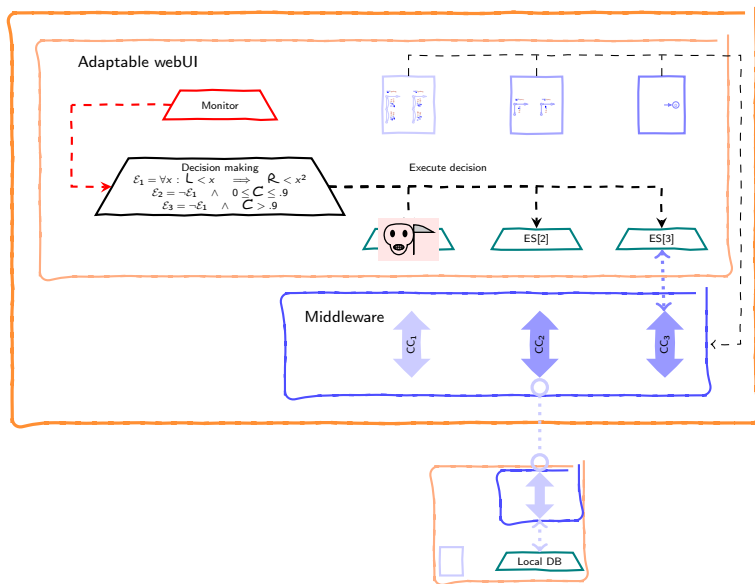


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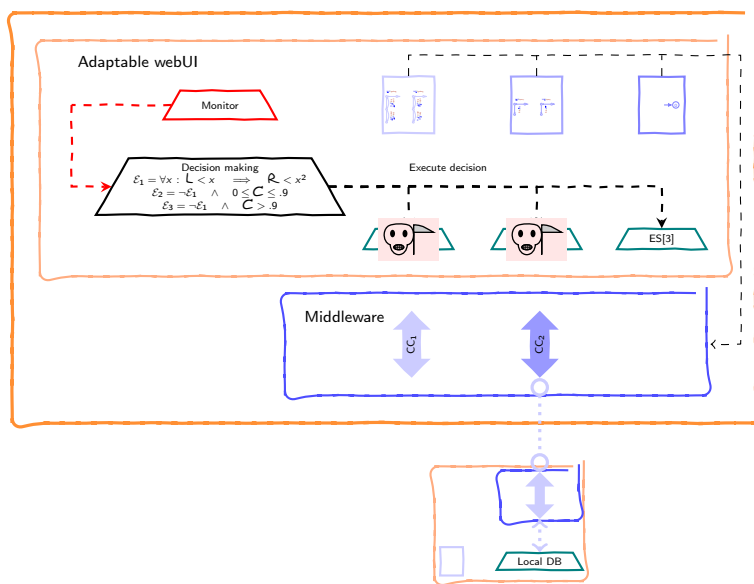
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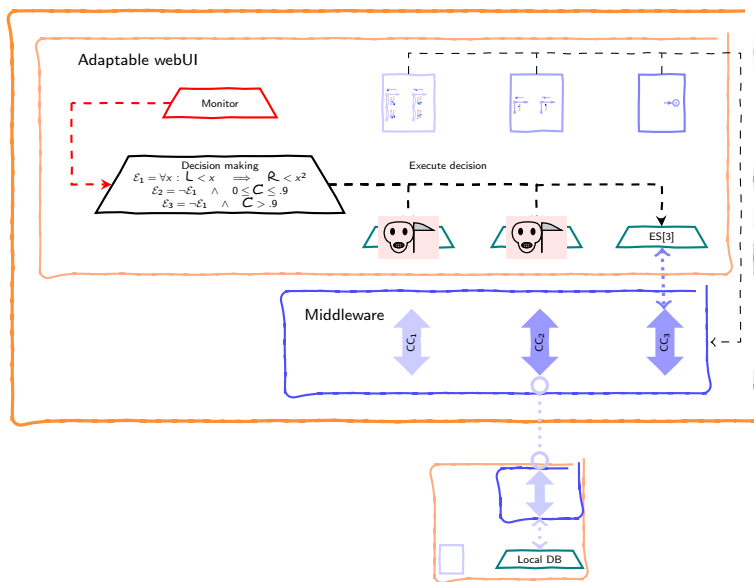
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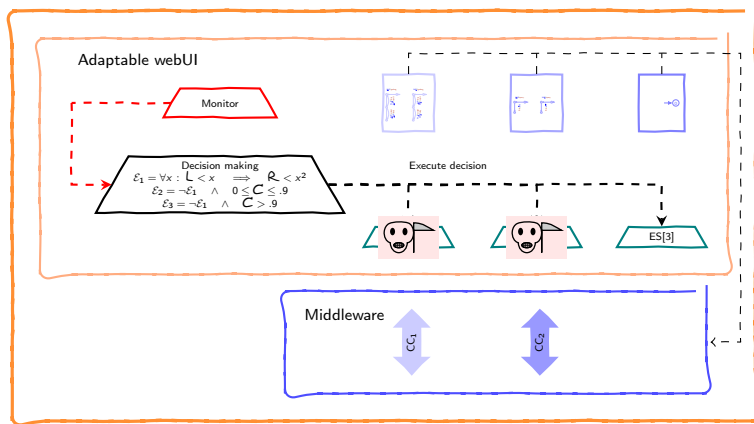


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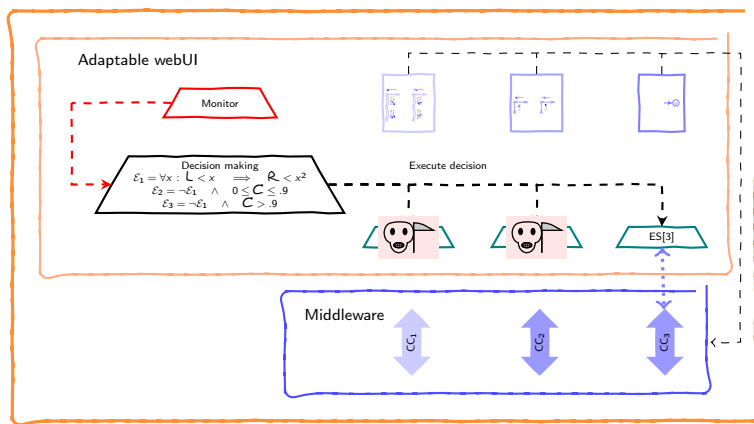


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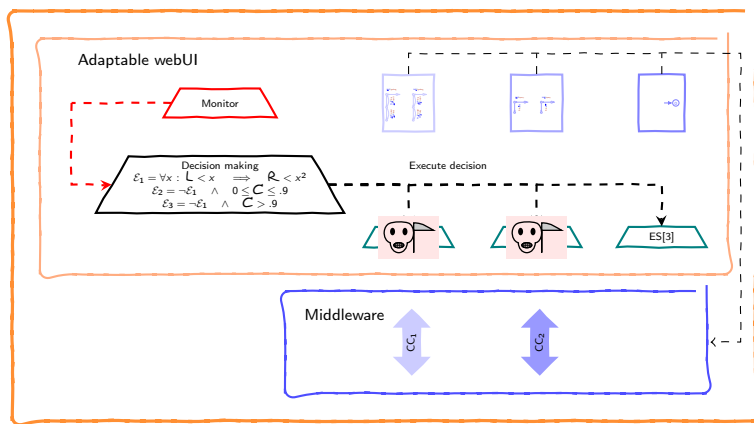
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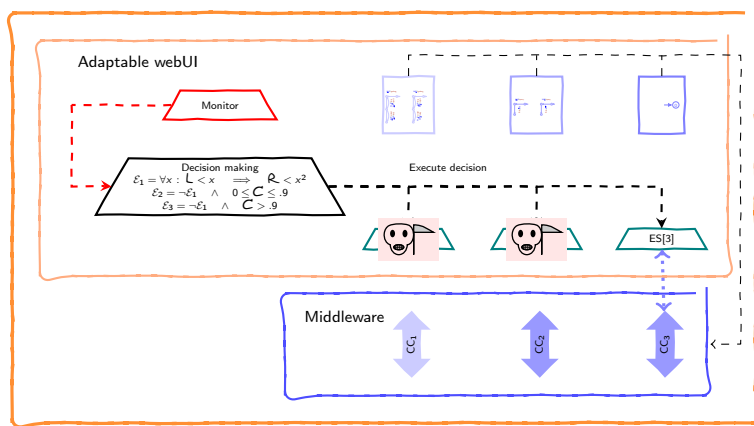
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– Epilogue –

Wrapping up

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- ▶ Take a look at the paper for
 - ▶ considerations about the control loop
 - ▶ details about adaptation of the running example

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- ▶ more general combinators (e.g., monitor passing data to execution scenarios)
- ▶ abrupt termination
 - ▶ move to graceful adaptation (e.g., ensuring transactional behaviour)

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R3: Presentation

- Apologies (it was a rush)...thank for the great help in improving the presentation

Thank you!

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